

# COSC 1100



Class Exercise



# COSC 1100

## Class Exercise



User Interfaces: Planning a User Interface  
Part 1 of 2



# User Interfaces: Planning a User Interface

## **Goal of Part 1:**

Experience planning for the design a program with UI.

## Requirements

- Did you come prepared?
  - Watched the UI with Tkinter video series?
  - Especially the ones about planning and accessibility?
- Collaboration.
  - You can work alone or with 1-2 colleagues during this exercise. (i.e. groups of 1-3)
  - The group will share their planning ideas at the end of this class, which will contribute to your grade.

## How to Plan a Design

- What types of things can happen when someone starts creating a design for an application without planning in advance?
- What information is required?
- Note the provided design template...  
(same used in the videos)

## Problem Description

- Space Eaters, a space efficiency consulting firm, requires a GUI-based application that will accept from a user the length and width of a rectangle and then allow for the user to initiate the calculation and display of the area and perimeter of that rectangle.
- Additionally the user should have some way to reset the form to its original state, prior to entry of a length and width as well as a clear way to close the form and end the application.
- Space Eaters is looking for a design that will integrate well into their current Windows environment with regard to style, colour, helpfulness and keyboard access.

## User Interface Planning – Requirements

We are in **planning**, but your program's design has to include the following:

- Use grid geometry manager to define rows and columns.
- There should be a title for the window, e.g. “Space Eaters” or “Rectangle Calculator”.
- The window should have a defined size (geometry).
- An appropriate number of controls to accept all required input and display all required output.
- Labels to indicate what input and output widgets are for
- Buttons that will execute all required functionality.
- Accessibility considerations: tab order, tooltips, hotkeys...

## User Interface Planning – What To Submit

- Based on the problem description, prepare a document that includes the following:
  - A visual representation of what your program would look like, in a grid-based format.
  - A list of expected widgets that will be rendered on the window...
    - For each widget, specify its name, content, location, and any other specific properties that must be set.
  - Pseudocode for the “Calculate” operation.
- Submit this document to DC Connect (**COSC 1100 -> Activities -> Assignments -> Class Exercise: User Interfaces**), including all group members' names if you are working in a group



## Summary

- Supposing you watched the whole video series about creating a user interface with Python...
  - How will the properties that you have identified help you write code for your user interface?
  - Can you estimate how many lines of code you will require to create your UI?

# COSC 1100

## Class Exercise



User Interfaces: Creating a UI with Tkinter  
Part 2 of 2



# User Interfaces: Creating a UI with Tkinter

## Goal of Part 2:

Experience developing the GUI for a program and implementing some simple functionality.

## Requirements

- Did you come prepared?
  - Reviewed this week's content including the User Interfaces in Python (tkinter) video series.
  - Completed the Pre-Class Activity for the week.
- Collaboration.
  - Once again, while you've got access to plans and you can feel free to discuss, *everyone will do their own coding today!*
  - Each student must submit their finished application to earn full marks on this week's Class Exercises.

## Problem Description (Review)

- Space Eaters, a space efficiency consulting firm, requires a GUI-based application that will accept from a user the length and width of a rectangle and then allow for the user to initiate the calculation and display of the area and perimeter of that rectangle.
- Additionally the user should have some way to reset the form to its original state, prior to entry of a length and width as well as a clear way to close the form and end the application.
- Space Eaters is looking for a design that will integrate well into their current Windows environment with regard to style, colour, helpfulness and keyboard access.

## Reviewing Some Potential Plans for this GUI

- Let's look at some of the submitted plans...  
(or at least something related)

## User Interface for Space Eaters

- Earlier this week, we developed a visual and written plan for the creation of a user interface that will solve the Space Eaters problem.
- Now we are going to write the tkinter code that represents the finished user interface.
- Starter code that includes the necessary functions should be provided for you to use!
- By the end of this exercise, the form should be able to be run, with its content being complete, readable, and accessible.
- The buttons should also do what is required.

## User Interface Coding – What to Submit

- Based on a plan from earlier this week and using the starter code file if you would like, complete the Python code file that includes the window and generates all required widgets.
  - Besides commenting, whitespace, and the window declaration this will be 2-3 lines of code per widget.
  - It might sound like a lot but many of them will be almost the same and there are no control structures or anything complex like that.
- Test it out! Does it work?
- Submit this code file to DC Connect (**COSC 1100 -> Activities -> Assignments -> Class Exercise: User Interfaces**)



## Discussion

- Testing was mentioned here!
  - How do we test something “properly”?
  - How can you tell if your application code is working?